



Theory of explosion limits of hydrogen/oxygen mixtures

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ABSTRACT

Analytical expressions for the classical Z-shaped pressure-vs-temperature explosion limit curve of hydrogen/oxygen mixtures are expeditiously derived based on the leading-order, algebraically linear reactions between the (H, HO₂, H₂O₂) radicals and the (H₂, O₂) background reactants. The analysis yields a detailed ignition mechanism (DIM) and then a skeletal ignition mechanism (SIM) describing the Z-curve, both of which result in close agreement with those computationally obtained with the original reaction mechanism. The solution leads to the ready identification of the first-second and second-third quadratic limits, the associated lower and upper turning points, and the single first, second, and third limits, with the concomitant demonstration that the conventional second limit, $[M] = 2k_1/k_9$, represented by the classical crossover-temperature, is inadequate to describe the transition between the first and third limits. A curvature-reversal, inflection point embedded within the second limit is identified, leading to the precise indication of the controlling transition chemistry between the low- and high-pressure regimes.

1. Introduction

The ability to predict and describe the ignition state of a combustible is essential for the design of combustion devices as well as the control of unwanted fires and explosions. Among various combustibles, the hydrogen/oxygen mixture exhibits a characteristic Z-shaped explosion limit curve in the pressure-vs-temperature plot, which clearly demonstrates the nonmonotonic nature of the global fuel reactivity. As shown in Fig. 1, as the pressure increases from a low value within a specific temperature range, the mixture undergoes transition from a non-explosion state to an explosion state, then back to a non-explosion state, and finally to an explosion state again, resulting in three transition boundaries respectively designated as the first, second and third explosion limits. These limits in turn constitute a lower turning point and an upper turning point, which respectively indicate the transition from the first limit to the second limit, and then from the second limit to the third limit.

The reaction mechanism controlling these three limits is largely understood in that, as pressure increases from a low value: (1) the chain branching cycle involving the H, O and OH radicals overwhelms the surface termination reactions, forming the first limit; (2) the increasing importance of the three-body reaction $H+O_2+M \rightarrow HO_2+M$ (designated

as R9, see Table 1) with increasing pressure generates the weakly reactive radical HO₂, which terminates the overall reaction and forms the second limit; and (3) with further increasing pressure, the higher concentration of HO₂ leads to its re-activation, with the formation of H₂O₂ and subsequently the OH and H radicals, constituting a high-pressure branching cycle and hence the third limit. The characteristic Z-curve can be readily reproduced through computation with detailed reaction mechanism.

Extensive efforts have been conducted to derive explicit analytical expressions for the various limits constituting the Z-curve, either individually, or collectively consisting of one or both turning points. Specifically: (1) the quadratic first-to-second limits, incorporating the lower turning point, is reported in [1]; (2) the second limit, without the turning points, is well known as the classical crossover temperature of the H₂/O₂ system [2]; (3) the third limit, without the upper turning point, is reported in [3]; and (4) the comprehensive Z-curve, encompassing all the three limits including the lower and upper turning points, is reported in [4,5], utilizing eigenvalue analysis. The recent work of [6] on effects of H₂ doping on the ignition of CO/O₂ mixtures extends the controlling reactions considered in [4,5] by incorporating the additional participation of H₂O₂.

The analysis of [6] has invoked several assumptions, as follows. First,

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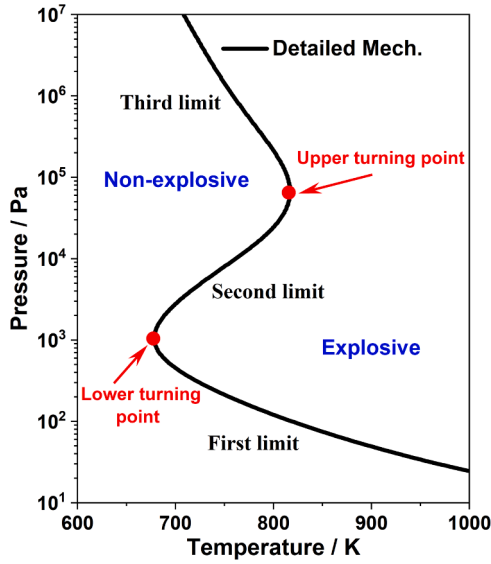


Fig. 1. The Z-curve response for the explosion limits of H_2/O_2 mixtures; calculation based on specifications in Section 4.

because of the relatively small concentrations of the radicals in the combustion mixture during the ignition transient, the bulk thermodynamic state of the mixture in terms of its temperature, pressure and concentrations of the reactants H_2 and O_2 are assumed to be unaffected by the participating reactions. Consequently, ignition is induced through, and defined by, the runaway of the radicals H , O , OH , HO_2 and H_2O_2 . Second, during ignition, the O and OH radicals are approximated as quasi-steady, leaving H , HO_2 and H_2O_2 as the only chain carriers whose evolution need to be tracked. Third, the controlling reactions then just involve these specific radicals with the bulk reactants H_2 and O_2 which are assumed to be of the given, fixed concentrations. Consequently, each individual reaction rate varies solely with the concentration of a *single*, participating radical and hence is *algebraically linear*. Fourth, the only noticeable quantitative deviation from predictions based on the linear reactions arises from the *algebraically nonlinear* reaction (R5): $\text{HO}_2 + \text{HO}_2 \rightarrow \text{H}_2\text{O}_2 + \text{O}_2$, which involves two HO_2 radicals.

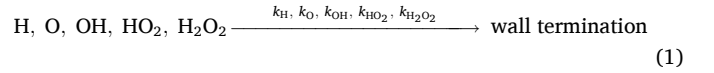
second-limit expression. The study then identifies, for the first time, a curvature-reversal, *inflection point* located on the second limit, which precisely distinguishes the transition from the first limit to the third limit through the low- and high-pressure mechanisms. This then demonstrates that the second limit is just the *transition phase* between the first and third limits instead of being an independent limit by itself.

In the following we shall first present the Z-curve analysis based on linear chemistry, leading to the reduced and then the skeletal ignition mechanisms. This is followed by the derivation and understanding of the various limits and the state of the inflection point.

2. Formulation of reduced ignition mechanism (RIM)

Following the above discussion, Table 1 lists the adopted first-order linear reactions, together with the nonlinear reaction (R5) whose influence has been demonstrated to be small [7], with the associated kinetic parameters abstracted from [8]. Similar results can be obtained by using the reaction rate parameters from alternate mechanisms.

The surface reactions of the chain carriers are the following:



with the corresponding reaction rate constant per unit volume given by $k = (1/2)\varepsilon\bar{v}(S/V)$, where $\bar{v} = (8k_{\text{B}}T/\pi m)^{1/2}$ represents the average velocity of the chain carriers at temperature T , m is the molar mass, S and V the surface area and volume of the spherical chamber, respectively, and k_{B} the Boltzmann constant. The sticking coefficient ε measures the destruction efficiency upon collision with the wall, with values ranging from 10^{-5} to 10^{-2} for glass and quartz.

By considering only the five radicals, H , O , OH , HO_2 and H_2O_2 , the conservation equations are:

$$\frac{d}{dt} \begin{bmatrix} [\text{H}] \\ [\text{O}] \\ [\text{OH}] \\ [\text{HO}_2] \\ [\text{H}_2\text{O}_2] \end{bmatrix} = J_5 \begin{bmatrix} [\text{H}] \\ [\text{O}] \\ [\text{OH}] \\ [\text{HO}_2] \\ [\text{H}_2\text{O}_2] \end{bmatrix} + \begin{bmatrix} I \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad (2)$$

where $I = k_4[\text{H}_2][\text{O}_2]$ is the rate of the initiation reaction (R4): $\text{H}_2 + \text{O}_2 \rightarrow \text{H} + \text{HO}_2$, and the associated 5×5 Jacobian matrix is:

$$J_5 = \begin{bmatrix} -k_1[\text{O}_2] - k_9[\text{O}_2][\text{M}] - k_{\text{H}} & k_2[\text{H}_2] & k_3[\text{H}_2] & k_{17b}[\text{H}_2] & 0 \\ k_1[\text{O}_2] & -k_2[\text{H}_2] - k_{\text{O}} & 0 & 0 & 0 \\ k_1[\text{O}_2] & k_2[\text{H}_2] & -k_3[\text{H}_2] - k_{\text{OH}} & 0 & 2k_{15}[\text{M}] \\ k_9[\text{O}_2][\text{M}] & 0 & 0 & -k_{17b}[\text{H}_2] - k_{\text{HO}_2} & 0 \\ 0 & 0 & 0 & k_{17b}[\text{H}_2] & -k_{15}[\text{M}] - k_{\text{H}_2\text{O}_2} \end{bmatrix} \quad (3)$$

However, since this reaction involves the product of two HO_2 molecules of low concentration, its participation is expected to have only a small, secondary effect on the reacting mixture prior to the onset of full-scale ignition, and thus only slightly affects the temperature and pressure of the upper turning point, as demonstrated in [7].

Recognizing the essential nature of the above assumptions, the first major objective of the present study is to quantitatively demonstrate their adequacy, which leads to an *expeditious* and straightforward derivation of a *reduced ignition mechanism* (RIM) for the entire Z-curve. Subsequently, by further assessing the importance of the individual participating terms, the RIM is simplified to a *skeletal ignition mechanism* (SIM), with clear indication of the role of the individual terms.

Using the SIM, the second major objective is to derive and understand the various explosion limits. In particular we shall demonstrate the essential inadequacy of, and subsequently revise, the conventional

It is noted that for an H_2/O_2 mixture with and without N_2 as diluent, we have $[\text{M}] = [\text{H}_2] + [\text{O}_2] (+ [\text{N}_2])$, and can express $[\text{O}_2] = f_{\text{O}_2}[\text{M}]$, $[\text{H}_2] = f_{\text{H}_2}[\text{M}]$. By assuming ideal gas, the total gas concentration $[\text{M}]$ is directly proportional to pressure, as $[\text{M}] = P/R^0T$.

Using the CSP (computational singular perturbation) analysis of Ref. [9] along the explosion limit curve, the eigenvalues of the local Jacobian can be used to rank the modes as FAST (with large and negative eigenvalues) or EXPLOSIVE (with positive eigenvalues). For each mode assumed to be fast, the species with the larger radical pointer value is considered a candidate as a quasi-steady state (QSS) species. Due to the large temperature and pressure range of the explosion limit solution, the range of the eigenvalues of matrix J_5 is also large. Consequently, different criteria of the limiting time scale need to be used to determine whether a certain mode is fast, and the first step is to compare

Table 1

Reactions for H_2/O_2 system. Units are $\text{cm}^3\text{-mol-s-cal-K}$, and $k = AT^n \exp(-E_a/RT)$.

No.	Reaction	A	n	E_a
R1	$\text{H} + \text{O}_2 \xrightarrow{k_1} \text{O} + \text{OH}$	1.040E+14	0.00	15286.0
R2	$\text{O} + \text{H}_2 \xrightarrow{k_2} \text{H} + \text{OH}$	5.080E+04	2.67	6292.0
R3	$\text{OH} + \text{H}_2 \xrightarrow{k_3} \text{H} + \text{H}_2\text{O}$	4.380E+13	0.00	6990.0
R4	$\text{H}_2 + \text{O}_2 \xrightarrow{k_4} \text{H} + \text{HO}_2$	5.176E+05	2.43	53502.0
R5	$\text{HO}_2 + \text{HO}_2 \xrightarrow{k_{5,1}} \text{O}_2 + \text{H}_2\text{O}_2$	1.300E+11	0.00	-1630.0
	$\text{HO}_2 + \text{HO}_2 \xrightarrow{k_{5,2}} \text{O}_2 + \text{H}_2\text{O}_2$	3.658E+14	0.00	12000.0
R9	$\text{H} + \text{O}_2(+\text{M}) \xrightarrow{k_9} \text{HO}_2(+\text{M})$	4.650E+12	0.44	0.00
	$k_{9,\infty}$	1.737E+19	-1.23	0.00
R15	$\text{H}_2\text{O}_2(+\text{M}) \xrightarrow{k_{15}} \text{OH} + \text{OH}(+\text{M})$	2.000E+12	0.90	48749.0
	$k_{15,\infty}$	2.490E+24	-2.30	48749.0
R17b	$\text{H}_2 + \text{HO}_2 \xrightarrow{k_{17b}} \text{H}_2\text{O}_2 + \text{H}$	7.150E+04	2.54	21404.0

the eigenvalues of the matrix under different conditions, as shown in Fig. 2(a). Since one of the five eigenvalues is always positive, only the absolute values of the four negative eigenvalues are shown in Fig. 2(a). It is then seen that under low pressure conditions, the absolute values of eigenvalues 1 and 2 are significantly larger than the other two eigenvalues, and when the pressure rises to around 1 atmosphere, eigenvalue 3 gradually approaches eigenvalues 1 and 2. Thus under low pressure conditions, there are two fast modes, while under high pressures, there are three fast modes. Fig. 2(b) then shows the radical pointers with the species for the cumulative projections on the top three modes, indicating that the radical pointers of the OH and O radicals are close to unity in the entire pressure range, while that of H is a small number under low pressures, but approaches unity at high pressures. Consequently, throughout the entire explosion limit range, OH and O can be treated as QSS species, while H can only be treated as QSS under high pressure conditions.

With the above understanding, the conservation equations for the three participating radicals, H, HO₂ and H₂O₂, are given by:

$$\frac{d}{dt} \begin{bmatrix} [\text{H}] \\ [\text{HO}_2] \\ [\text{H}_2\text{O}_2] \end{bmatrix} = J_3 \begin{bmatrix} [\text{H}] \\ [\text{HO}_2] \\ [\text{H}_2\text{O}_2] \end{bmatrix} + \begin{bmatrix} I \\ I \\ 0 \end{bmatrix} \quad (4)$$

with the associated 3×3 matrix being:

$$J_3 = \begin{bmatrix} 2k_1[\text{O}_2] - k_9[\text{O}_2][\text{M}] - k_H & k_{17b}[\text{H}_2] & 2k_{15}[\text{M}] \\ k_9[\text{O}_2][\text{M}] & -k_{17b}[\text{H}_2] - k_{\text{HO}_2} & 0 \\ 0 & k_{17b}[\text{H}_2] & -k_{15}[\text{M}] - k_{\text{H}_2\text{O}_2} \end{bmatrix} \quad (5)$$

For neutral conditions, the determinant of J_3 is:

$$F_3(T, P) = a[\text{M}]^4 + b[\text{M}]^3 + c[\text{M}]^2 + d[\text{M}] + e = 0 \quad (6)$$

with the coefficients being:

$$\begin{aligned} a &= 2f_{\text{O}_2}f_{\text{H}_2} \\ b &= \frac{2k_1f_{\text{O}_2}f_{\text{H}_2}}{k_9} - \frac{k_{\text{HO}_2}f_{\text{O}_2}}{k_{17b}} \\ c &= \frac{2k_1k_{\text{H}_2\text{O}_2}f_{\text{O}_2}f_{\text{H}_2}}{k_9k_{15}} + \frac{2k_1k_{\text{HO}_2}f_{\text{O}_2}}{k_9k_{17b}} - \frac{k_{\text{HO}_2}k_{\text{H}_2\text{O}_2}f_{\text{O}_2}}{k_{15}k_{17b}} - \frac{k_{\text{H}}f_{\text{H}_2}}{k_9} \\ d &= \frac{2k_1k_{\text{HO}_2}k_{\text{H}_2\text{O}_2}f_{\text{O}_2}}{k_9k_{15}k_{17b}} - \frac{k_{\text{HO}_2}k_{\text{H}}}{k_9k_{17b}} - \frac{k_{\text{H}_2\text{O}_2}k_{\text{H}}f_{\text{H}_2}}{k_9k_{15}} \\ e &= -\frac{k_{\text{H}}k_{\text{HO}_2}k_{\text{H}_2\text{O}_2}}{k_9k_{15}k_{17b}} \end{aligned}$$

The solution of Eq. (6), henceforth designated as the reduced ignition mechanism (RIM), as well as the corresponding detailed ignition mechanism (DIM), F_5 , for the analysis of J_5 , respectively yields the

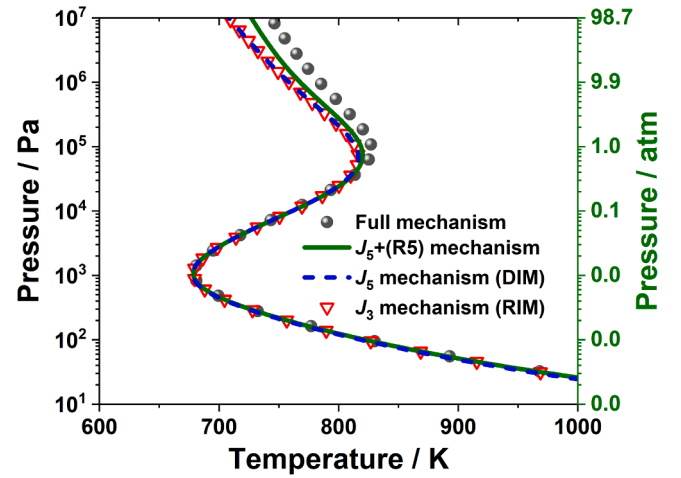


Fig. 3. Explosion limits with different reaction mechanisms.

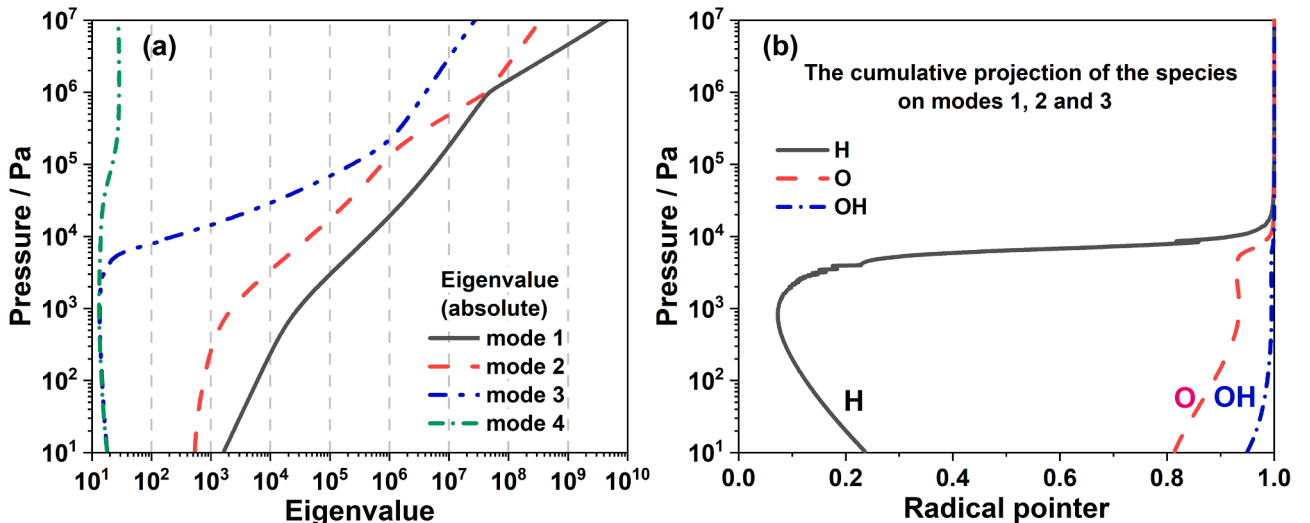


Fig. 2. The eigenvalues of each mode and radical pointers of the CSP radicals along the explosion limit curve.

general Z-curve response, with and without assuming QSS of O and OH.

Fig. 3 compares the Z-curve response for H_2/O_2 mixtures under various levels of assumption: (1) the numerical solution using the complete reaction mechanism of [8], with the residence time of 10 s; (2) the solution of J_5 (DIM), together with reaction (R5) [7]; (3) the solution of J_5 ; and (4) the solution of J_3 (RIM). The results indicate that: (i) the linear chemistry formulation renders the mixture to be slightly less explosive for the third limit including the upper turning point, which is

$$F_{SMM}(T, P) = 2 \left(\frac{k_9}{k_H} \right) \left(\frac{k_{15}}{k_{H2O2}} \right) \left(\frac{k_{17b}}{k_{H2O2}} \right) f_{O2} f_{H2} [M]^4 + \left(\frac{2k_1}{k_H} \right) \left(\frac{k_{17b}}{k_{H2O2}} \right) f_{O2} f_{H2} [M]^2 - \left(\frac{k_9}{k_H} \right) f_{O2} [M]^2 + \left(\frac{2k_1}{k_H} \right) f_{O2} [M] - 1 = 0 \quad (9)$$

primarily caused by the omission of the nonlinear reaction (R5); and (ii) the QSS assumption for O and OH is justified with the close agreement between J_3 and J_5 . These findings support the assumptions invoked in [7]. Consequently, we shall now proceed to derive an algebraically simpler and more transparent solution (SIM) including the various limiting situations. We note in passing that while the present study has been conducted for both H_2/O_2 and H_2 /air mixtures, only results from the H_2/O_2 mixtures are presented herein as understandings gained are the same for both mixtures.

3. Skeletal ignition mechanism (SIM)

To assess the dominant reactions constituting the RIM, sensitivity analysis was conducted for each coefficient in Eq. (6) by deleting one term at a time and comparing the computed results with those of the RIM. The terms in the coefficients of a , b , c , d and e in Eq. (6) are labeled as a , b_1 , b_2 , c_1 , c_2 , c_3 , c_4 , d_1 , d_2 , d_3 and e , with the second numeral, for example the numeral “1” in b_1 , indicating the first term in the coefficient b in Eq. (6).

Fig. 4 shows the sensitivity results of the individual terms in Eq. (6). It is evident that only five out of the eleven coefficients have noticeable impact on the explosion limit, namely:

$$a : 2f_{O2}f_{H2}, \quad c_1 : \frac{2k_1k_{H2O2}f_{O2}f_{H2}}{k_9k_{15}}, \quad c_3 : \frac{k_{H2O2}k_{H2O2}f_{O2}}{k_{15}k_{17b}}, \quad d_1 : \frac{2k_1k_{H2O2}k_{H2O2}f_{O2}}{k_9k_{15}k_{17b}}, \quad e : -\frac{k_Hk_{H2O2}k_{H2O2}}{k_9k_{15}k_{17b}} \quad (7)$$

The above results can be explained as follows. First, we normalize Eq. (6) by the coefficient “ e ”, leading to:

$$F_3(T, P) = 2 \left(\frac{k_9}{k_H} \right) \left(\frac{k_{15}}{k_{H2O2}} \right) \left(\frac{k_{17b}}{k_{H2O2}} \right) f_{O2} f_{H2} [M]^4 + \left(\frac{2k_1}{k_H} \right) \left(\frac{k_{15}}{k_{H2O2}} \right) \left(\frac{k_{17b}}{k_{H2O2}} \right) f_{O2} f_{H2} [M]^3 - \left(\frac{k_9}{k_H} \right) \left(\frac{k_{15}}{k_{H2O2}} \right) f_{O2} [M]^3 + \left(\frac{2k_1}{k_H} \right) \left(\frac{k_{17b}}{k_{H2O2}} \right) f_{O2} f_{H2} [M]^2 + \left(\frac{2k_1}{k_H} \right) \left(\frac{k_{15}}{k_{H2O2}} \right) f_{O2} [M]^2 - \left(\frac{k_9}{k_H} \right) f_{O2} [M]^2 - \left(\frac{k_{15}}{k_{H2O2}} \right) \left(\frac{k_{17b}}{k_{H2O2}} \right) f_{H2} [M]^2 + \left(\frac{2k_1}{k_H} \right) f_{O2} [M] - \left(\frac{k_{15}}{k_{H2O2}} \right) [M] - \left(\frac{k_{17b}}{k_{H2O2}} \right) f_{H2} [M] - 1 = 0 \quad (8)$$

Two observations can be made from Eq. (8). First, each reaction rate is scaled, and hence directly affected, by that of the surface termination reaction affecting it, for example $k_1[O_2]$ and $k_9[O_2][M]$ by k_H , and

$k_{17b}[H_2]$ by k_{H2O2} . This is physically reasonable and clarifying. Second, except for the coefficient “ a ”, all the neglected terms, namely (b_1 , b_2 , c_2 , c_4 , d_2 , d_3), depend on reaction (R15) through ($k_{15}[M]/k_{H2O2}$). This is again reasonable, indicating that reaction (R15) can be neglected except for high-pressure situations scaled with $[M]^4$.

Consequently, based on the above analysis, $F_3(T, P)$ can be simplified to only five terms, expressed as a polynomial in $[M]$ and hence the system pressure:

which will be referred to as the skeletal ignition mechanism (SIM). The general yet simple and organized nature of Eq. (9) facilitates further analysis and physical understanding of the ignition response, presented next.

4. The ignition limits

4.1. The high- and low-pressure quadratic limits

For high-pressure (HP) situations, we remove the 0th-order term in Eq. (9) to yield F_{HP} :

$$F_{HP} = \frac{2k_9f_{O2}k_{15}k_{17b}f_{H2}}{k_{H2O2}k_{H2O2}} [M]^3 + \left(2k_1f_{O2} \frac{k_{17b}f_{H2}}{k_{H2O2}} - k_9f_{O2} \right) [M] + 2k_1f_{O2} = 0 \quad (10)$$

For low-pressure (LP) situations, we can readily remove the first, $[M]^4$, term in Eq. (9). Furthermore, since the first term in the coefficient for $[M]^2$ is related to the reactivation of the HO_2 radical as the system approaches the upper turning point, it can be neglected for the low-pressure situations. Consequently, the low-pressure quadratic limit is given by F_{LP} :

$$F_{LP} = -k_9f_{O2} [M]^2 + 2k_1f_{O2} [M] - k_H = 0 \quad (11)$$

Fig. 5 shows close agreement between SIM and the high- and low-pressure quadratic limits.

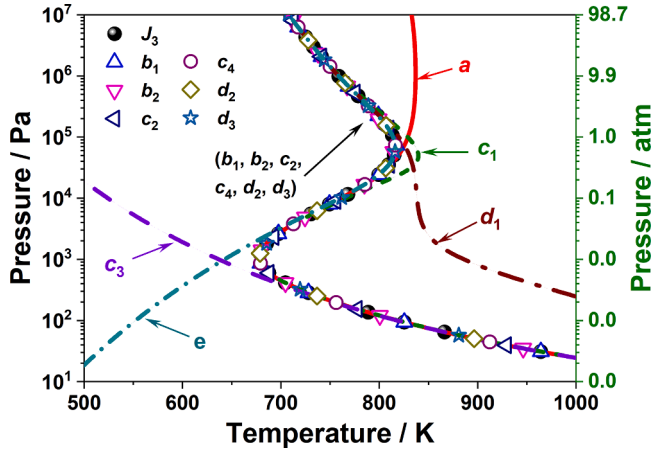
4.2. The upper and lower turning points

With the above expressions for the high- and low-pressure quadratic limits obtained, namely Eqs. (10) and (11), the upper and lower turning points of the explosion limit curve can be readily derived. Specifically, since each turning point is given by the condition: $(\partial T / \partial P) = (\partial F / \partial P) / (\partial F / \partial T) = 0$, and since $(\partial F / \partial T) \neq 0$, we have $(\partial F / \partial P) = 0$ at the turning points, which then yields the respective solutions for the turning points:

$$\frac{\partial F_{LP}}{\partial P} = -k_9[M] + k_1 = 0 \quad (12)$$

$$\frac{\partial F_{HP}}{\partial P} = 3 \frac{2k_9k_{15}k_{17b}f_{H2}}{k_{H2O2}k_{H2O2}} [M]^2 + \frac{2k_1k_{17b}f_{H2}}{k_{H2O2}} - k_9 = 0 \quad (13)$$

The calculated upper and lower turning points and their specific locations on the explosion limit curve are shown in Fig. 5 (as well as Fig. 1), demonstrating that they are well captured. These two turning points are important features of the Z-shaped hydrogen explosion curve,

Fig. 4. Sensitivity analysis of individual terms in $F_3(T, P)$.

being the parameters that anchor the lower and upper explosion peninsulas. For the lower turning point, it is controlled by the competition of (R1), (R9) and the wall elimination of H, with k_9 and k_H accounting for the H radical removal rate and k_1 the chain branching rate. For the upper turning point, it is controlled by (R1), (R9), (R15), (R17b) and the wall termination of HO_2 and H_2O_2 .

4.3. The individual single limits

Knowing the high- and low-pressure quadratic limits, we next identify the three individual limits. Thus, the first limit is obtained by neglecting the second-order, $[M]^2$, term in the low-pressure quadratic limit, Eq. (11), to yield:

$$[M] = k_H / 2k_1 f_{O_2} \quad (14)$$

Similarly, the third limit is obtained by neglecting the 0th-order term in the high-pressure quadratic limit, Eq. (10), to yield:

$$2k_9 k_{15} k_{17b} f_{H_2} [M]^2 + (2k_1 k_{17b} f_{H_2} k_{H_2O_2} - k_9 k_{H_2O_2} k_{H_2O_2}) = 0 \quad (15)$$

Identification of the second limit is however more subtle. Conventionally, it is given by:

$$[M] = 2k_1 / k_9 \quad (16)$$

which is the well-known classical expression for the crossover temperature [10]. While it is perceptively simple, it does not provide the

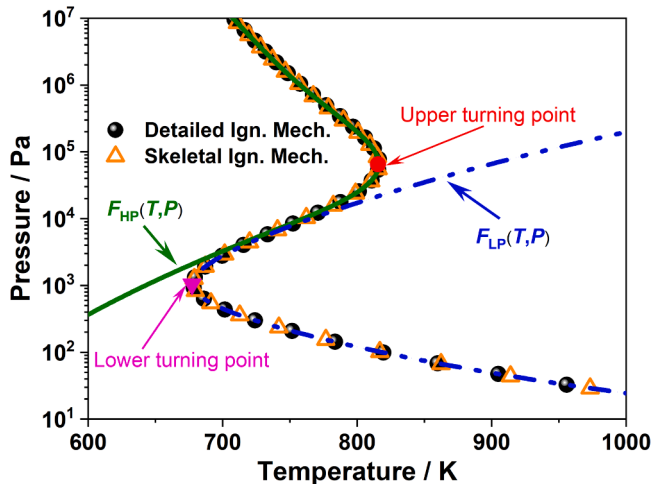
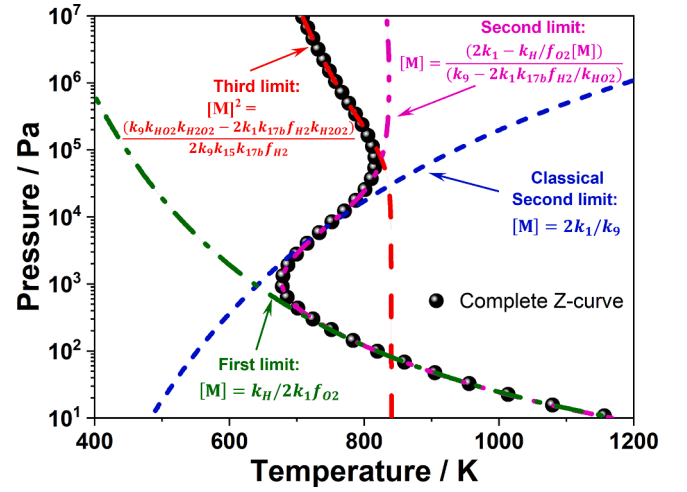
Fig. 5. Comparison of explosion limits determined for all limits, the high-pressure limit, F_{HP} , and the low-pressure limit, F_{LP} .

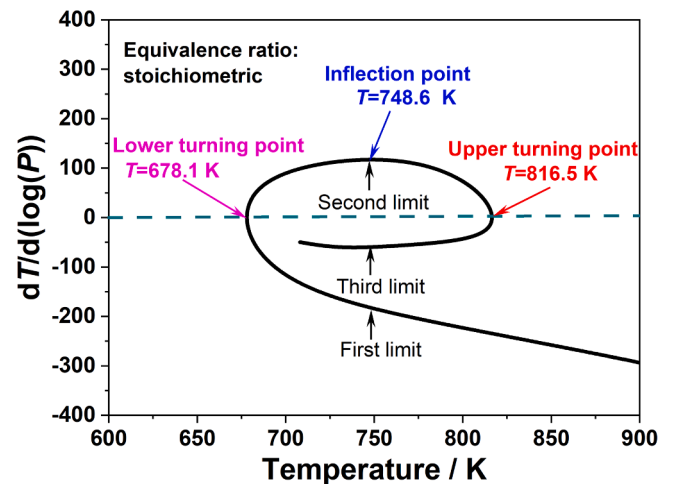
Fig. 6. Explosion with the single limits.

connection with the first and third limits and will necessarily become inaccurate and break down near both turning points. As such, a more realistic second limit should exhibit the proper transition towards these two limits, and needs to be identified from the complete SIM, Eq. (9), instead of the low- or high-pressure quadratic limits. An inspection of Eq. (9) then shows that while we can readily neglect the highest order, $[M]^4$, term since the second limit is at lower pressures, the remaining four terms need to be retained. That is, while the third and fourth terms in Eq. (9) constitute the original second limit, Eq. (16), and are independent of the third and first limits, the second and fifth terms in Eq. (9) would respectively provide the transition to these two limits. As such, the revised second limit is given by,

$$[M] = \left(2k_1 - \frac{k_H}{f_{O_2} [M]} \right) / (k_9 - 2k_1 k_{17b} f_{H_2} / k_{H_2O_2}) \quad (17)$$

Eq. (17) degenerates to Eq. (16) only with the simultaneous neglect of the second terms in its numerator and denominator, hence losing the connectivity to the first and third limits. Eq. (17) can then be designated as either the second limit or the consolidated second limit, while Eq. (16) as the conventional second limit.

Fig. 6 compares the explosion limits described by the three single limits, Eqs. (14), (17), and (15), together with the classical, conventional second limit and the result obtained from the complete solution, Eq. (9), demonstrating good agreement for the limit solutions.

Fig. 7. Critical states of the explosion limits and the inflection point, determined from J_3 .

The simplicity of the SIM allows the clear identification of the leading-order reactions governing each individual limit. That is, the first limit, Eq. (14), is controlled by reaction (R1): $\text{H} + \text{O}_2 \rightarrow \text{O} + \text{OH}$, and is influenced by the oxygen concentration and the wall termination of H. The *conventional* second limit, Eq. (16), is determined by the ratio of the key branching to termination reaction rates of the $\text{H} + \text{O}_2$ reactions, which are k_1 and k_9 . The *consolidated* second limit, Eq. (17), incorporates the suppression effects on k_1 with wall termination reaction of H, as well as the inhibitory effect on the chain termination reaction through reactions (R17b) and (R1), where the partially generated HO_2 from (R9) is converted into H, O, and OH radicals. It also accounts for the wall termination of HO_2 , and the concentrations of the reactants $[\text{O}_2]$ and $[\text{H}_2]$, providing the proper transitions. The third explosion limit, Eq. (15), is controlled by reactions (R9), (R17b) and (R15), and is influenced by the H_2 concentration and the HO_2 and H_2O_2 wall termination reactions. These are the simplest expressions for the three explosion limits of H_2/O_2 mixtures.

5. The inflection point: transition state between low- and high-pressure regimes

In discussions of the Z-curve explosion limits, it is frequently suggested that the conventional second limit delineates the transition of the explosion mechanism controlled by the low-pressure $\text{H}-\text{O}_2$ chemistry to that controlled by the high-pressure $\text{HO}_2-\text{H}_2\text{O}_2$ chemistry. Fundamentally, since this description of the (P, T) transition state is *uniform* along the entire classical second limit, it must necessarily break down at the lower and upper turning points. In fact, it is also obvious that the second limit must possess a *curvature reversal* to allow its transition between the first and third limits in the P -vs- T plot.

This state of transition is characterized by variation of the gradient dT/dP , shown as $dT/d(\log P)$ in Fig. 7, in the range of the second limit between the lower and upper turning points. It is then clear that while the gradient varies between the vanishing states of the two turning points, as demonstrated earlier, its *own gradient* vanishes at a specific state between them, through an *inflection point*. This inflection state, characterizing the reversal in curvature, then indicates the transition from low- to high-pressure explosion chemistries.

We next study the significance of the inflection point in distinguishing the controlling chemistry along the second limit. Utilizing CSP analysis, the radical with the largest radical pointer in the explosive mode, characterized by the largest positive eigenvalue, is identified as the dominant radical controlling explosion. Fig. 8 shows the changing trend of the dominant radicals along the explosion limit curve, with Fig. 8(a) showing the location of the dominant radicals on the explosion limit curve and Fig. 8(b) indicating the corresponding values of the radical pointers. It is then seen that the inflection point is situated at the trade-off regime of the radical pointers of H and HO_2 on the second explosion limit. Furthermore, the H radical dominates the first explosion limit, while the HO_2 radical dominates the third explosion limit. As pressure increases, the H pointer first increases, and then rapidly decreases beyond the lower turning point. Subsequently, H_2O_2 replaces H as the second most important radical, with its pointer increases with pressure. In contrast, the O pointer remains relatively small and is passed by HO_2 upon crossing the lower turning point. Then the HO_2 pointer rapidly increases while the H pointer decreases. The inflection point occurs where the H pointer attains a small value, say 0.1, while the HO_2 pointer concurrently assumes a large value of 0.9. As the H_2O_2 radical emerges, the HO_2 pointer is already near unity. Subsequently, the H_2O_2 pointer increases while the HO_2 pointer decreases; they eventually converge to approximately 0.5, indicating equal importance. These findings demonstrate that the inflection point, and the state of curvature reversal of the second limit, represents a sharp transition separating the low- and high-pressure chemistries in the H_2/O_2 explosion limits. The transition occurs between H- and HO_2 -related reactions, highlighting the role of the inflection point in delineating distinct chemical pathways in the explosion limits.

We finally assess the dependence of the inflection point on the state of the mixture; noting that the inflection point can be readily determined from any of the Z-curve solutions: namely Eq. (8), Eq. (9), or the second limit Eq. (17). Fig. 9 shows the pressure and temperature of the inflection point for different equivalence ratios, demonstrating that the inflection point has the minimum temperature at the state of stoichiometric concentration, indicating that it is the most favorable state for the transition from the low- to the high-pressure chemistries involving the production of HO_2 , as expected.

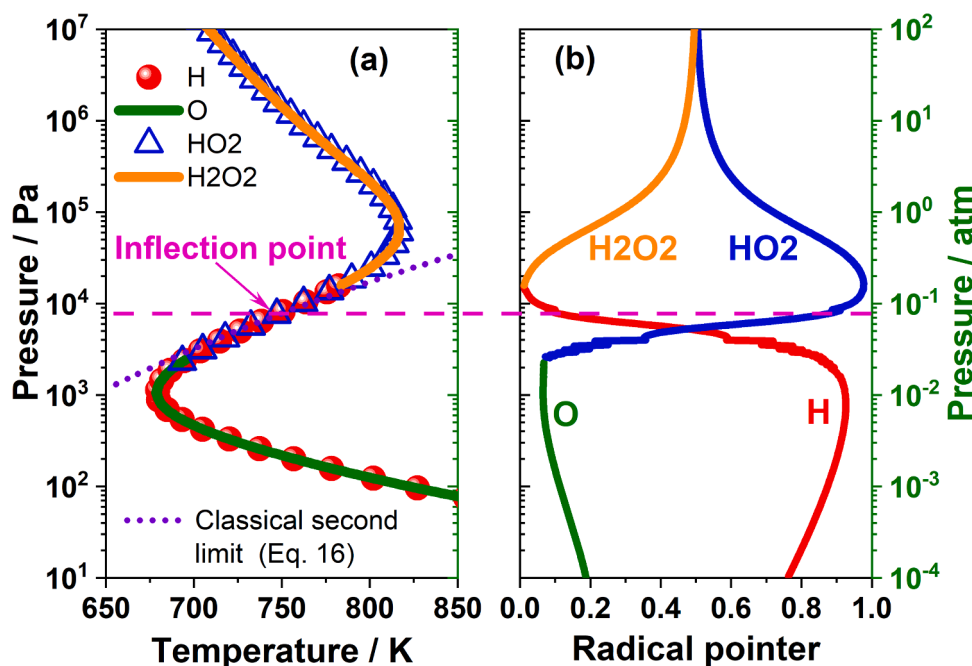


Fig. 8. The leading two species with the largest radical pointers for the explosive mode. (a) Positions of the radical pointers. (b) Magnitudes of the dominant radical pointers.

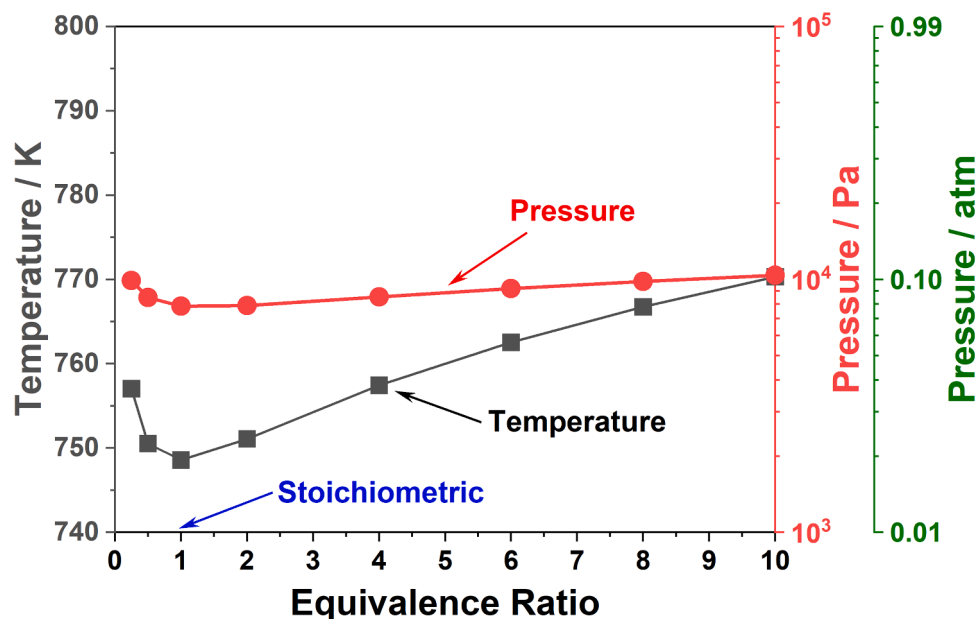


Fig. 9. State of pressure and temperature of the inflection point.

5. Concluding remarks

In the present study we have presented an expeditious derivation of the classical Z-shaped hydrogen/oxygen explosion limit, leading to explicit specifications of the entire Z-curve, the specialized first-second and second-third quadruple limits including the respective lower and upper turning points, and the individual first, second and third limits. These expressions identify the controlling mechanisms for the respective transitions, particularly the revised expression of the second limit. The study also recognizes the (P , T) state of an inflection point along the consolidated second limit that precisely separates the pressure-temperature explosion space into the low- and high-pressure regimes, and thereby suggests that the consolidated second limit simply represents the *nonmonotonic* transition between the first and third limits instead of having its own independent characteristics.

Together with previous contributions, it is suggested that the classical problem of the hydrogen/oxygen explosion limits is largely analytically solved, with explicit and clarified descriptions of the various limits and changeover states. In terms of practical utility, the analytical expressions derived herein not only enhance the understanding of the explosion phenomena, but they can also be used for assessment in various applications. For example, the expressions of the limits can be readily grafted unto large computation programs to significantly reduce the effort needed to simulate the states of ignition.

Finally, the concept of linear chemistry holds substantial potential in analyzing chemically reacting systems through the organization and simplicity in the formulation, and merits further exploration.

Novelty and significance

Hydrogen, being a foundational fuel, exhibits a highly nonlinear ignition response in the form of a Z-shaped pressure-vs-temperature explosion limit. However, a comprehensive theoretical analysis of the limits and the transition between them has not been adequately conducted for this important classical problem. The present study derives a simplified yet accurate expression for the *entire* explosion limit, which leads to the identification of the various limits and the transition turning points. The conventional second limit is considered to be inadequate and

consequently revised to provide the proper transition between the first and third limits. An inflection state due to curvature reversal on the second limit is also identified, yielding precise transition for ignition from low- to high-pressure chemistry. The study completes the theory of hydrogen/oxygen explosion limits and provides guidance for hydrogen utilization and safety.

CRediT authorship contribution statement

Jie Liu: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Wenkai Liang:** Investigation. **Chung K. Law:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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